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## ME 306P: Solid Mechanics Laboratory

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| <b>Course Code</b>   | ME 306P                    |
| <b>Course Name</b>   | Solid Mechanics Laboratory |
| <b>L-T-P-C</b>       | 3-0-0-3                    |
| <b>Prerequisites</b> | Nil                        |
| <b>Approval</b>      | 5th Senate (Oct 2013)      |
| <b>Discontinued</b>  | yes                        |

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### Course Contents

- Comparison of the properties of ductile and brittle materials.
- Tensile test to obtain stress strain curves, young's modulus, 0.2% proof stress,
- Compression tests to compare the failure behaviour of the ductile and brittle materials.
- Shear test.
- Three point bend test for bending strength.
- Determination of Young's modulus of a material by bending test.
- Buckling load for columns with different end conditions.
- Evaluation of spring index and calculation of proof stress for compression spring.
- Impact tests and comparison of energies for ductile and brittle materials.
- Torsion tests to obtain modulus of rigidity and fracture behaviour of brittle materials subjected to torsion.
- Hardness tests. Empirical relations of hardness and strength.
- Fatigue test.
- Surface roughness tests to understand the different surface roughness obtained by various manufacturing processes.